

3(a)	Combine resistance of X and Y = $\frac{(2.0)(3.0)}{2.0 + 3.0} = 1.2 \, \Omega$ Combine resistance of circuit = $1.8 + 1.2$ $= 3.0 \, \Omega$ Current = $\frac{1.5}{3.0}$ $= 0.50 \, \text{A}$
3(b)	ratio = $\frac{I^2(1.2)}{I^2(3.0)}$ $= 0.40$
3(c)	New combine resistance of X and Y = $\frac{(4.5)(3.0)}{4.5 + 3.0} = 1.8 \, \Omega$ New ratio = $\frac{I^2(1.8)}{I^2(1.8 + 1.8)}$ $= 0.50$ The ratio increased as resistance of X is increased. Current through internal resistance and the combined resistance of X and Y are the same. As resistance of X increases, potential difference across X and across Y increases. Hence the power dissipated across X and Y increased. Hence ratio increased.
4(a)	By Fleming's left hand rule, the moving charge experiences a magnetic force which is always perpendicular to its velocity and the magnetic flux density.